

Valtti Järvi

FOUNDATIONAL TIME SERIES MODELS FOR EMERGENCY DEPARTMENT OCCUPANCY FORECASTING

Bachelor's Thesis
Faculty of Information Technology and Communication Sciences
Examiner: M.Sc. Eetu Pulkkinen
November 2025

ABSTRACT

Valtti Järvi: Foundational time series models for emergency department occupancy forecasting
Bachelor's Thesis
Tampere University
Bachelor's Programme in Computing and Electrical Engineering
November 2025

Emergency department crowding compromises patient safety and operational efficiency. Accurate occupancy forecasting enables proactive resource allocation. This thesis evaluates whether foundational time series models can accurately forecast emergency department occupancy in a zero-shot setting. This capability would eliminate the computational costs and manual effort required to train traditional machine learning models.

This thesis compares five open-source foundational models (TimesFM, Chronos, Sundial, Moirai, and TiRex) against state-of-the-art baseline machine learning models using emergency department occupancy data from Tampere University Hospital. The foundational models are evaluated precisely on the same data as state-of-the-art models were evaluated for direct comparison. The thesis assessed predictive performance, the impact of historical context length and external covariates, and computational requirements.

The results show that univariate foundational models, particularly TimesFM, TiRex, and Sundial, outperform best previous baseline when provided with sufficient historical context. Increasing context length consistently improved accuracy up to maximum possible context length of 12,840 hours. However, Moirai, the only model capable of processing multivariate data, failed to benefit from external covariates. Computationally, all models demonstrated attractive inference times and memory usage. These findings suggest that zero-shot foundational models offer a viable, scalable alternative to traditional forecasting methods for Emergency department crowding management.

Keywords: time series forecasting, emergency department occupancy forecasting, time series foundational models, foundation models

The originality of this thesis has been checked using the Turnitin OriginalityCheck service.

TIIVISTELMÄ

Valtti Järvi:
Kandidaatin tutkielma
Tampereen yliopisto
Tieto- ja sähkötekniikan kandidaattiohjelma
Marraskuu 2025

Päivystyspoliklinikoiden ruuhkautuminen vaarantaa potilasturvallisuuden ja heikentää toiminnan tehokkuutta. Tarkka kuormituksen ennustaminen mahdollistaisi resurssien kohdentamisen ennakoivasti. Tässä työssä arvioidaan, pystyvätkö esikoulutetut aikasarjamallit (engl. foundational time series models) ennustamaan päivystyksen potilasmääriä ilman mallin erillistä kouluttamista historiallisen potilasdatan avulla (engl. zero-shot). Tämä kyvykkyys poistaisi perinteisten koneoppimismallien kouluttamiseen vaadittavat laskennalliset kustannukset ja manuaalisen työn.

Tässä työssä vertaillaan viittä avoimen lähdekoodin esikoulutettua mallia (TimesFM, Chronos, Sundial, Moirai ja TiRex) suorituskyvyltään edistyneimpiin koneoppimismalleihin käyttäen Tampereen yliopistollisen sairaalan päivystyksen kuormitusdataa. Vertailukelpoisuuden saavuttamiseksi esikoulutetut mallit arvioidaan käyttämällä samaa dataa kuin vertailumallien arvioimiseen käytettiin. Tutkimuksessa arvioitiin ennustetarkkuutta, historiatietojen pituuden ja ulkoisten muuttujien vaikutusta, sekä laskennallisia vaatimuksia.

Tulokset osoittavat, että yhden muuttujan perusmallit, erityisesti TimesFM, TiRex ja Sundial, suoriutuvat paremmin kuin paras aiempi vertailumalli, kun niille syötetään riittävästi historiallista aikasarjadataa. Aikasarjadataan pidentäminen paransi tarkkuutta johdonmukaisesti aina suurimpaan mahdolliseen 12 840 tunnin pituuteen asti. Ainoa monimuuttujadataa tukeva malli, Moirai, ei kyennyt hyödyntämään ulkoisia muuttujia. Laskennallisesti kaikkien mallien laskenta-ajat ja muistinkäyttö ovat kohtuullisella tasolla nykyaikaisille tietokoneille. Tulokset viittaavat siihen, että esikoulutetut aikasarjamallit tarjoavat varteenotettavan ja skaalautuvan vaihtoehdon perinteisille ennustusmenetelmille päivystyksen ruuhkautumisen hallinnassa.

Avainsanat: aikasarjaennustaminen, Päivystyspoliklinikoiden kuormituksen ennustus, esikoulutetut aikasarjamallit, perustamallit

Tämän julkaisun alkuperäisyys on tarkastettu Turnitin OriginalityCheck -ohjelmalla.

USE OF AI IN THESIS

I have utilized AI tools in my thesis:

☐ No

☒ Yes

The AI tools utilised in my thesis and their purposes are described below:

Names and versions of AI tools: Gemini 3, Gemini 2.5 Pro, Claude Sonnet 4.5, Grok 4

Purpose of using AI tools: All AI tools were used to enhance the writing process by improving grammar, style, and clarity. Gemini 2.5 Pro was used to create tables and equations. Claude Sonnet 4.5 was used for coding tasks during the production of experimental results.

Sections where AI tools were used: All AI tools listed were used throughout the thesis.

I acknowledge that I am fully responsible for the entire content of my thesis, including the parts generated by AI, and accept accountability for any violations of ethical standards in publications.

PREFACE

I want to express my gratitude to my supervisor M.Sc. Eetu Pulkkinen for his helpful guidance and support throughout this process. I am also grateful to Dr. Jalmari Tuominen for his invaluable advice, particularly for insights into emergency department processes. I would like to thank them both for providing this exciting thesis topic. I wish to acknowledge CSC – IT Center for Science, Finland, for computational resources. Finally, I would like to thank my family for their meaningful support and encouragement.

Tampere, 20th November 2025

Valtti Järvi

CONTENTS

1. Introduction	1
2. Background	3
2.1 Emergency department	3
2.1.1 Definition and consequences of crowding	3
2.1.2 Emergency department forecasting	4
2.2 Time series forecasting	4
2.3 Foundation models	5
2.3.1 Pre-training and fine-tuning paradigm	6
3. Methodology	8
3.1 Data	8
3.2 Experimental setup	9
3.3 Evaluation	10
3.4 Models	11
3.4.1 TimesFM	11
3.4.2 Chronos	11
3.4.3 Sundial	12
3.4.4 Moirai	13
3.4.5 TiRex	13
3.4.6 Benchmark models	14
4. Results and Discussion	16
4.1 Predictive performance	16
4.2 Effect of context length	17
4.3 Effect of covariates	18
4.4 Computational cost	19
4.5 Limitations	21
5. Conclusion	22
6. References	24
Appendix A: Computational costs for multivariate forecasts	28

1. INTRODUCTION

Emergency department (ED) crowding negatively impacts a range of clinically relevant patient outcomes [1], including higher mortality rates [2], [3] and longer lengths of stay [4]. Accurate predictions of future ED occupancy levels would enable proactive resource allocation aiming to alleviate or even prevent future crowding events.

Emergency department forecasting refers to the process of predicting the future status of the ED, typically using timeseries forecasting models. Traditionally, researchers have trained these using historical use data, often incorporating external variables such as weather and traffic data [5], [6]. However, training machine learning models presents several challenges. First, every hospital has unique characteristics. A model trained with data from one hospital may not accurately predict occupancy levels for other hospitals. Consequently, each hospital requires a new model at the time of implementation and repeatedly throughout the production cycle, which can be a computationally heavy and time-consuming task. Second, accessing the required data for each hospital can be a challenge due to either regulatory or practical reasons.

Foundation models (FMs) are artificial intelligence (AI) systems that can be used in a wide range of different tasks and are trained on massive amounts of diverse data [7]. In the context of emergency department forecasting, they offer a set of unique advantages in comparison to traditional machine learning algorithms. First, there is no need for model training. Second, they can be deployed with minimal local data collection. Third, their massive size and training data allows them to discover patterns that traditional machine learning model could have missed. [7]

Foundation models do not need to be trained on task-specific data. If they can predict ED occupancy accurately enough, the difficult task of model training can be skipped. This could save a substantial amount of computation and man-hours.

This thesis evaluates whether the newly introduced foundational time series models can be used to predict ED occupancy of Tampere University Hospital ED "Acuta." A public dataset from a GitHub repository [8] containing the total hourly occupancy of the ED from June 20, 2018 to June 19, 2019 is used as the target variable, and the performance of the FMs is evaluated in comparison to the state-of-the-art (SOTA) models documented by Tuominen et al. [5]. They used exactly the same dataset which makes direct comparison possible.

This thesis addresses the following research questions: (1) how does the performance of the FMs compare to the traditional SOTA models; (2) are there differences between the performances of the existing FM implementations, and (3) how do the FMs differ in terms of computational requirements?

The thesis is structured as follows. Chapter 2 provides the necessary background on emergency departments, crowding, time series forecasting, and foundation models. Chapter 3 details the experimental setup and the dataset used. Chapter 4 presents the empirical results and their implications in the context of existing knowledge. Finally, Chapter 5 provides final conclusions that can be drawn from the work.

2. BACKGROUND

This chapter provides the background for forecasting ED crowding. We first outline the role of EDs and clarify what crowding means and why it matters. We then introduce forecasting in the ED context and review core time series concepts used in this work. Finally, we present foundational time series models and explain how they differ from traditional machine learning models in terms of training and deployment for specific applications.

2.1 Emergency department

EDs deliver medical care to patients at any time with a wide range of acute conditions and injuries [9]. Swift diagnostic evaluation and timely medical or surgical interventions are provided for patients requiring immediate care [1].

2.1.1 Definition and consequences of crowding

Emergency departments (EDs) serve a critical role in healthcare systems by providing immediate access to medical services [1]. Worldwide, they function as a crucial 'safety net,' addressing acute health problems and responding to local and national disasters [10], [11]. In the United States, more than half of all inpatient hospital admissions originate from the ED [10].

ED crowding does not have a single widely accepted definition. The proposed frameworks include National Emergency Department Overcrowding Score (NEDOCS) and Emergency Department Work Index (EDWIN) [1]. Regardless of the definition, crowding is always

a situation in which demand for service exceeds the ability to provide care within a reasonable time, causing physicians and nurses to be unable to provide quality care

as was expressed in the *Canadian Journal of Emergency Medicine* [12].

Regardless of the definition, crowding has become a worldwide public health issue [1]. The negative impacts can be divided into three broad categories based on their target: the patient, the staff, and the entire healthcare system. First, crowding has been repeatedly associated with increased mortality [2] and increased number of medication errors [13].

Second, crowding has been shown to increase the stress levels of nurses and physicians [14]. Third, crowding results in ambulance diversions diverting ambulances to another ED [10], which delays patient access to care and disrupts coordination between hospitals. In addition, crowding compromises the ability of the ED to respond to mass emergencies or at least their capability to provide quality care to less urgent patients [15].

2.1.2 Emergency department forecasting

Emergency department forecasting refers to the process of predicting future occupancy of the ED. Accurate forecasting would enable proactive resource allocation that could ultimately prevent crowding along with its harmful consequences [9]. Previous literature investigates the predictability of a wide set of target variables [9]. These include amount of patient arrivals [16], amount of patient admissions [17], length of stay [18], and occupancy [5].

Almost all previous ED forecasting literature has used either *statistical models* or *machine learning models*. Statistical models include ARIMA variants and exponential smoothing methods [19]. Machine learning models include artificial neural networks (ANNs), decision trees, support vector machines, and regression-based approaches such as linear, logistic, and Poisson regression [9], [20].

On the other hand, the use of foundation models has not been reported in the previous literature for ED occupancy data specifically. However, FMs have been reported to get promising results in other domains like energy load forecasting [21] and restaurant sales forecasting [22]. This suggests that foundation models could be a promising tool for ED forecasting.

In practice, forecasting relies on historical observations to learn recurring patterns such as daily, weekly, and seasonal cycles, as well as long-term trends. Explanatory variables, such as weather, traffic, and hospital bed availability can be used to improve accuracy. Forecasts can be point estimates or full predictive distributions. [5]

2.2 Time series forecasting

The goal of time series forecasting is to predict the future evolution of a data sequence based on its past values. External data sequences can be included for more accurate forecasts [6]. For example, future hourly ED occupancy can be predicted using past ED occupancy data, calendar information, and weather.

A discrete-time time series is a set of discrete values $\{x_t\}$, where t denotes a specific time [19]. In this thesis, the term 'time series' refers always to discrete-time time series. The intervals between t are usually fixed and of equal length.

The term univariate refers to data that consists of a single attribute or variable. In contrast, multivariate data consists of multiple related variables observed over time, where the relationships between variables can provide additional information for forecasting. It can be denoted as $\{\mathbf{x}_t\}$, where $\mathbf{x}_t$ is a vector. Other variables than the one we predict are called covariates or exogenous inputs. [19]

For a time series to be suitable for forecasting, it must exhibit identifiable patterns. The trend component represents the long-term direction or gradual change in the data, which, in the case of ED occupancy data, may reflect factors such as population growth. Seasonality means repeating patterns that occur at regular intervals. For example, ED occupancy in Acuta peaks daily between 16:00-19:00 and is at its lowest between 04:00-07:00 [5]. Random noise refers to irregular and random fluctuations that cannot be explained by any pattern. After all, many events, such as individual accidental injuries, occur randomly. These and many other patterns, including patterns from covariates, make forecasting possible. [19]

A *model* is a mathematical equation that describes the patterns found in the data. The objective is to find a model that effectively captures the underlying structure of the time series. This relationship can be formally expressed with a general equation:

$$y_{t+1} = f(y_t, y_{t-1}, \dots, \mathbf{x}_{t+1}, \mathbf{x}_t, \dots) + \epsilon_{t+1} \quad (2.1)$$

The function f represents the model itself. It specifies the relationship between the target variable y_{t+1} and the past values of the target variable $y_t, y_{t-1}, \dots$, and the present and past values of the covariates $\mathbf{x}_t, \mathbf{x}_{t-1}, \dots$. The term ϵ_t is the random noise that is not explained by the model. The function f can be a simple statistical model or a more complex machine learning model like a neural network. [6]

The future is always unpredictable to some extent. Therefore, predicting the entire probability distribution of a future value, rather than just a single value, is often useful. This is called a forecast distribution. Prediction intervals represents a range of values that the future value could take given some probability. For example, a 90 % prediction interval contains the actual value with 90 % probability. In contrast, a point forecast represents the best single estimate of a future value. It is often the mean or median of the forecast distribution. [6]

2.3 Foundation models

Foundation models are artificial intelligence (AI) systems that perform well in a wide range of diverse tasks. These models are trained on massive amounts of diverse data. Foundation models use deep neural network architectures with parameters ranging from tens of millions to hundreds of billions, depending on the application domain. For example,

foundation models have been developed for time series forecasting. A single foundation time series model can predict diverse time series data, from ED occupancy to stock prices. [7]

Foundation models differ from traditional task-specific machine learning models. Traditional models are typically trained from scratch for a single specific task, such as forecasting ED occupancy for one particular hospital. In contrast, foundation models are pre-trained on massive amounts of diverse data and can be adapted to multiple different tasks. Advancements in computing have made it possible to train such large models for multiple different purposes [7].

Transfer learning is a concept of transferring knowledge from one learned task to other tasks [23]. From vast training data, fundamental structures that are true for all tasks in one domain, are learned [23]. For example, ED occupancy and stock prices could have same kind of daily fluctuation. Transfer learning and the vast size of foundation models is what makes them work [7].

In Bommasani et al. [7], the authors state three advancements that have enabled the size of foundation models. First, improvement in computational hardware. Second, the Transformer architecture that efficiently used this hardware for training [24]. Third, access to larger datasets for training.

2.3.1 Pre-training and fine-tuning paradigm

Deep learning is a part of machine learning that uses deep neural networks to learn from data. These networks learn data representations at several levels. [25] Deep learning powers most modern AI models. All foundation models in this thesis use deep learning, as does the TFT baseline from Tuominen et. al [5] [26].

In Domingos [27], the author outlines three key aspects of developing machine learning models. First step is choosing representation, or model. In case of deep learning, this is done by choosing a neural network architecture. Then the model is optimized or trained to fit the data. Training is done by adjusting the model parameters to minimize a loss function. This step is computationally intensive and often requires massive datacenters and significant amount of time [7]. The final step is evaluation, where the model's performance is measured using data not seen during training. Deep learning follows this framework [25].

Foundation models use different framework and they represents a paradigm shift from traditional machine learning framework. For a specific task, already pre-trained foundation model can be selected. The foundation model is often fine-tuned for a specific task. After fine tuning, the model is evaluated. [7]

This approach saves a lot of computation, time and effort, since the foundation model is already pre-trained. This approach makes advanced AI models accessible to a much broader audience. In case of ED occupancy forecasting, every hospital has its own characteristics. Therefore model trained for one hospital would perform poorly for other hospitals. Training a separate model for each hospital would be expensive. Foundation models can solve this problem.

3. METHODOLOGY

This section details the experiments used to evaluate whether foundation time series models can accurately forecast ED occupancy. A direct comparison with the baseline models from Tuominen et al. [5] requires the use of identical data and evaluation metrics.

First, the ED occupancy dataset and external covariates are described. Second, the experimental setup for 24-hour zero-shot forecasting task and different context lengths and external covariate setups are described. Used hardware and measurements for computational costs are also introduced. Third, the evaluation metrics are defined. Finally, all five foundation models and four benchmark models are described. [5]

3.1 Data

Tuominen et al. [5] used ED occupancy as a forecasting metric. To directly compare results with Tuominen et al., this thesis uses ED occupancy as its crowding metric. All patients from all ED treatment spaces are summed up to this metric. All parts of the input-throughput-output model are affected in occupancy, which makes it useful. [5]

The ED occupancy data used in this thesis comes from Tampere University Hospital's (TAYS) electronic health record. TAYS is an academic, Level 1 trauma center in Tampere, Finland. When the study by Tuominen et al. [5] was conducted, TAYS served a primary population of 535,000 within the Pirkanmaa Hospital District and provided tertiary care for a total catchment population of 900,700. The hospital's combined emergency department, Acuta, managed an annual census of approximately 100,000 patients at that time. The ED had a total capacity of 111–118 patients, which included 70 beds (plus 7 reserve) and 41 seats for ambulatory patients. [5]

The ED occupancy data spans from January 1, 2017, to June 19, 2019. It consists of hourly registered ED visits. As by Tuominen et al. [5], a whole year of occupancy data from June 20, 2018, to June 19, 2019, was used as a testing dataset in this thesis. Tuominen et al. also used a total of 167 external covariates. All of the covariate categories are described in table 3.1. Covariates can be divided into categories of past and future covariates. Future covariates include data points for the time steps for which the forecasts are made. For example, weather forecasts and public events can be used as future covariates. [5]

Category	Description
Hospital beds	Hourly measurements of available beds across 24 healthcare facilities within the catchment region.
Trends	Hourly Google Trends intensity for the keyword <i>Acuta</i> .
Traffic	Hourly vehicle throughput recorded at 33 bidirectional monitoring stations in the Pirkanmaa region.
Weather	Ten historical meteorological variables observed at the nearest weather station.
Public events	Daily counts of organized public events within the Tampere area.
Website visits	Web analytics for two hospital domains combined with Google search activity for "Acuta".
Calendar	Encodings of temporal attributes including weekday, month, national holidays, and associated lead-lag indicators.
Technical analysis (TA)	Autoregressive features derived from the target series, such as moving averages.
All categories (All)	Composite feature set of all 167 features comprising all categories.

Table 3.1. *Categories of External Covariates (adapted from Tuominen et al. [5])*

3.2 Experimental setup

The model's task is to predict hourly occupancy levels for the next 24 hours. For a one year testing dataset, this means 365 predictions. All of the predictions are grouped together and compared against the ground truth values from the testing dataset. These results are then directly compared with results from Tuominen et al. [5].

To evaluate univariate model performances across different historical window sizes, a set ten different context lengths were tested. These were grouped into temporal and fixed-token windows. The temporal contexts included calendar based intervals of 1 day, 3 days, 1 week, 2 weeks, 4 weeks, and 1 year. The maximum available window size from January 1, 2017 to June 19, 2018 spanned a total of 12840 hours or 535 days, was also included. To fill the gaps in between these window sizes, fixed-length windows of 512, 2048 and 4096 hours were also evaluated.

To test which kind of external covariates foundation models can leverage, the covariates were divided into subsets according to their categories. To enable a direct comparison

with Tuominen et al. [5], one forecasting was also performed with all covariates grouped together. Adding all the covariates dramatically increases the input dimensionality. This, in turn, rapidly scales the computational load especially with longer context lengths. This is why only context lengths 256, 512 and 1024 hours were tested.

Univariate forecasts were performed on a local workstation equipped with an Intel Core i5-8600K (6-core @ 4.3 GHz) CPU and 16 GB of DDR4 RAM. Multivariate forecasts were computed on the Puhti supercomputer provided by CSC - IT Center for Science, Finland. This experiment was run on one NVIDIA Volta V100 GPU with 32 GB of VRAM.

For evaluation computational requirements of each foundation model, time needed to make forecasts and peak RAM during inference were measured. Time it takes to generate forecasts is called *inference time*. Inference time was measured for the whole test set spanning one year. Peak RAM usage is measured over the whole inference run, including loading the model to the computer memory.

3.3 Evaluation

In this thesis same metrics are used as in Tuominen et al. [5]. They used mean absolute error (MAE) and root mean squared error (RMSE) for point forecasts.

MAE is defined as

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|, \quad (3.1)$$

where y_t is the ground truth value and $\hat{y}_t$ is the prediction made by the model. RMSE is defined as

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2}, \quad (3.2)$$

using the same notation.

MAE measures forecast accuracy in the same units as the data and treats all errors equally [6]. This makes it easy to interpret. A MAE of 5 means forecasts are off by 5 patients on average.

RMSE squares the errors before averaging, which gives more weight to larger errors [6]. This makes RMSE more sensitive to outliers and particularly useful when large errors are more harmful than small ones. In ED occupancy forecasting, this property is valuable because large prediction errors could lead to understaffing or overstaffing. This could allow crowding or wasted resources.

3.4 Models

This section describes the baseline models from Tuominen et al. [5] and the FMs selected for this study. The FMs are compared using their architecture and key forecasting parameters. The *context length* is the number of past time steps the model uses to make a prediction. The *forecasting horizon* is the number of future time steps it predicts. All selected models have promising zero-shot capabilities, meaning they can be used to make forecasts without any additional training.

The foundation models were selected based on three criteria: open-source availability, ease of use, and performance on the GIFT-Eval benchmark [28]. All chosen models have promising zero-shot capabilities and are available for download from GitHub. The models were downloaded from GitHub in September 2025. The exact model version names and links to their GitHub repositories are listed in Table 3.2. Median predictions were chosen over mean predictions for forecasts.

3.4.1 TimesFM

Developed by Google Research, *TimesFM* is a family of decoder-only transformer models designed for efficient forecasting [29]. The model uses a technique called patching. Patching divides the input time series into fixed-length segments. The model processes these segments instead of individual time points. TimesFM uses an input patch length of 32 time steps and an output patch length of 128 time steps. This reduces the number of steps required for long-horizon forecasting. Training minimizes the Mean Squared Error (MSE) loss on point forecasts.

The latest version, TimesFM 2.5 that was used in this thesis was released in September 2025. It has 200 million parameters and maximum context length of 16,384. The model was pretrained on approximately 100 billion time points from diverse sources. These included Google Trends, Wikimedia pageview statistics, and public datasets such as M4 and Electricity. This data was augmented with synthetic data from ARMA processes. As of September 2025, TimesFM 2.5 holds third place on the GIFT-Eval leaderboard [30].

3.4.2 Chronos

Chronos is a family of foundational time series models developed by Amazon Science [31]. It adopts a language modeling framework and treats time series forecasting as a next-token prediction task. Based on the T5 transformer architecture [23], Chronos processes continuous time series data by scaling and quantizing it into a fixed vocabulary of discrete bins. The model is trained using cross-entropy loss to predict the next token. This frames forecasting as a classification problem at each time step. To generate forecasts, the model

predicts one token at a time. Each predicted token is used as input for predicting the next one. These discrete tokens are then dequantized and inverse-scaled to produce continuous forecast values. The model samples multiple forecast paths to generate a prediction distribution. From this distribution it calculates the mean, median, and quantile forecasts.

In this thesis newer version *chronos-bolt-base* was used. It is a 205 million parameter variant of the Chronos family [32]. This model improves upon original architecture by extending the maximum context length from 512 to 2048, while maintaining maximum prediction horizon of 64 [33].

Chronos-bolt-base processes inputs and generates outputs differently than older versions of Chronos. It groups input time series into patches processing them as continuous vectors rather than discrete tokens. This allows model to handle longer context lengths more efficiently. Outputs are not generated autoregressively as in previous versions, but the whole forecast distribution is generated by one forward pass. [31]

The training data was augmented using the TSMIXup and KernelSynth algorithms. TSMixup creates new time series by combining existing ones. KernelSynth generates synthetic data using Gaussian processes. As of September 2025, *chronos-bolt-base*, holds fourth place in GIFT-Eval benchmark for models with training data overlap to the test set [30].

3.4.3 Sundial

Sundial was developed by Tsinghua University in February 2025 [34]. Defining characteristic of Sundial is that it outputs probabilistic predictions directly. This is done by producing multiple forecast samples, where mean median and quantile forecasts are calculated. The prediction mechanism is autoregressive at the patch level.

Sundial uses a decoder-only transformer architecture and works directly on continuous values rather than using quantization. The model processes time series data in patches of size 16. It is trained using a novel TimeFlow Loss function.

The version of Sundial used has a context length of 2880 but can handle shorter, arbitrary context lengths. The model was trained on TimeBench. This is a large-scale dataset created by the authors, which contains over a trillion data points from real-world sources like weather and finance. As of November 2025, Sundial holds the 10th place on the GIFT-Eval leaderboard [30].

3.4.4 Moirai

Developed by Salesforce AI Research, *Moirai* features a masked encoder transformer architecture designed specifically for heterogeneous, multivariate time series [35]. Its key feature is using varying patch sizes based on data frequency. Higher-frequency data uses larger patches. Each patch is processed by a linear layer and assigned a time ID and a variable ID. These IDs inject positional information into the transformer through rotary embeddings and learnable scalars. The transformer's output patches are projected to produce parameters for a mixture of probability distributions. These parameters include the weights for combining the distributions into a final weighted sum. This sum forms the final probabilistic forecast.

This study uses the *Moirai 2.0 small* variant, released in August 2025 [36]. Unlike its predecessor, this model utilizes a decoder-only transformer architecture with a single patch size, processing the input time series as a sequence of uniform patches. This design choice simplifies the input projection while retaining the ability to model complex temporal dependencies.

A key architectural distinction of Moirai 2.0 is its forecast generation mechanism. Instead of projecting outputs into a mixture of probability distributions, the model uses recursive multi-quantile decoding. This approach directly predicts the quantiles of the forecast distribution at each step, enhancing both probabilistic accuracy and inference speed.

The model contains approximately 12.4 million parameters, which is 96 % smaller than the original Moirai Large. It was trained on a robust corpus comprising the GIFT-Eval dataset, internal Salesforce operational data, and synthetic time series generated via the KernelSynth and TSMixup algorithms. As of September 2025, Moirai 2.0 small ranks sixth on the GIFT-Eval leaderboard [30].

3.4.5 TiRex

TiRex was developed by NXAI GmbH in collaboration with academic researchers [37]. It was released on May 2025. It is the only foundation model in this thesis based on the xLSTM architecture rather than a transformer architecture [38].

TiRex normalizes inputs using z-score normalization and processes data in patches of size 32. Input patches are projected into an embedding dimension by a residual block. This embedding is fed into the xLSTM. A second residual block projects the xLSTM output back into the patch dimension to produce forecasts. The outputs are scaled back to the original target space. The model predicts nine quantiles for each time step.

TiRex has approximately 35 million parameters, making it computationally efficient. Its xLSTM architecture propagates sequentially hidden states between patches. This removes

hard constraints on context length present in transformers. A context length of 2048 was used for training in the original study, but theoretically the model supports indefinitely large context lengths during inference.

The model’s training data comprises 47.5 million time series from Chronos’s training data, synthetic data similar to KernelSynth, and parts of the GiftEval pre-training dataset. As of November 2025, TiRex holds second place on the GIFT-Eval leaderboard among models with training data overlap to the test set [30].

Model	Version Name
Chronos ¹	chronos-bolt-base
TimesFM ²	timesfm-2.5-200m-pytorch
Moirai ³	moirai-2.0-R-small
TiRex ⁴	TiRex
Sundial ⁵	sundial-base-128m

Table 3.2. *Open-source models evaluated in this study.*

3.4.6 Benchmark models

The Seasonal Naïve (SN) model provides a baseline forecast by repeating the last observed value from the previous seasonal period. For a time series with a daily pattern, the forecast for a future time is simply the value observed at the same time one seasonal cycle ago (e.g., 24 hours prior). This approach is a standard benchmark for strongly seasonal data. [6]

ARIMAX (Autoregressive Integrated Moving Average with eXogenous inputs) is a statistical model that extends the standard ARIMA method. It combines ARIMA, which models the time series’s own autocorrelation structure, with a linear regression component. This regression component allows the model to incorporate the predictive information from external, or exogenous, variables. [6]

LightGBM (Light Gradient Boosting Machine) is a machine learning algorithm based on gradient boosting decision trees. It is recognized for its high efficiency and scalability, which makes it particularly effective for large-scale datasets. Tuominen et al. [5] applied LightGBM in their study forecasting ED occupancy, where it achieved the strongest performance among the models evaluated. [39]

The Temporal Fusion Transformer (TFT) is a deep neural network architecture designed specifically for multi-horizon time series forecasting [26]. It utilizes an attention-based encoder-decoder structure to learn complex temporal relationships at different scales. A key feature of TFT is its integrated interpretability, which allows for identifying significant

past time steps and important features.

4. RESULTS AND DISCUSSION

In this thesis, we establish four main findings about applying foundational models to emergency department occupancy forecasting. (1) Modern univariate FMs surpass a tuned gradient boosting baseline when they have long enough context length. (2) Increasing context length keeps improving accuracy, but the rate of improvement slows down with larger context lengths. (3) The multivariate covariates available for this study did not help the only FM that could consume them. (4) Lastly, Moirai delivers the best performance if computational costs are taken into account. The following sections elaborate on each finding.

4.1 Predictive performance

For each foundation model, the result from best performing context length setting was picked. For TimesFM, TiRex and Sundial it was the largest context length that was tested. For TimesFM and TiRex it was 12,840 hours and for Sundial 2048 hours. Moirai got its best result for second largest context length that was 8760 hours or one year. For Chronos, best result came using context length of 2048 hours. Best result using external covariates came with context length of 1024 but remained inferior to worst performing baseline model.

Table 4.1 shows that all univariate models except Chronos moderately outperformed best baseline model, LightGBM. Univariate models benefited from the large context length. Reason for poor performance of Chronos could be explained by it being the oldest model. [32] TimesFM outperformed other foundation models.

Model	FS	MAE	RMSE
Moirai	A	11.10	13.69
SN	U	9.53	12.58
ARIMA	U	7.78	10.50
Chronos	U	7.21	9.77
TFT	U	7.18	9.58
TFT	A	6.86	9.04
LightGBM	U	6.66	8.95
LightGBM	A	6.63	8.77
Sundial	U	6.57	8.85
TiRex	U	6.51	8.78
Moirai	U	6.35	8.59
TimesFM	U	<u>6.29</u>	<u>8.45</u>

Table 4.1. Best MAE and RMSE for each model. For each model, result from the best context length was picked. Foundational models are denoted with bold font. FS denotes Feature Set, where U is univariable and A is all covariates.

4.2 Effect of context length

Excluding Chronos, no significant differences in performance were observed between the FMs when using the same context length. Performance improves consistently when context length is increased. However, performance gains begin to diminish at context lengths beyond 2048 hours, as shown in Figure 4.1.

In Section 3.4, the content of training data for each model is introduced. Each model's training data consist of massive amounts of time series data, with substantial overlap in their use of public and well-known time series datasets. The fairly similar results could be explained by this overlap in model training data. A second possible explanation is that the models have captured all the patterns within the historical data, and the remaining residual component consists of random noise.

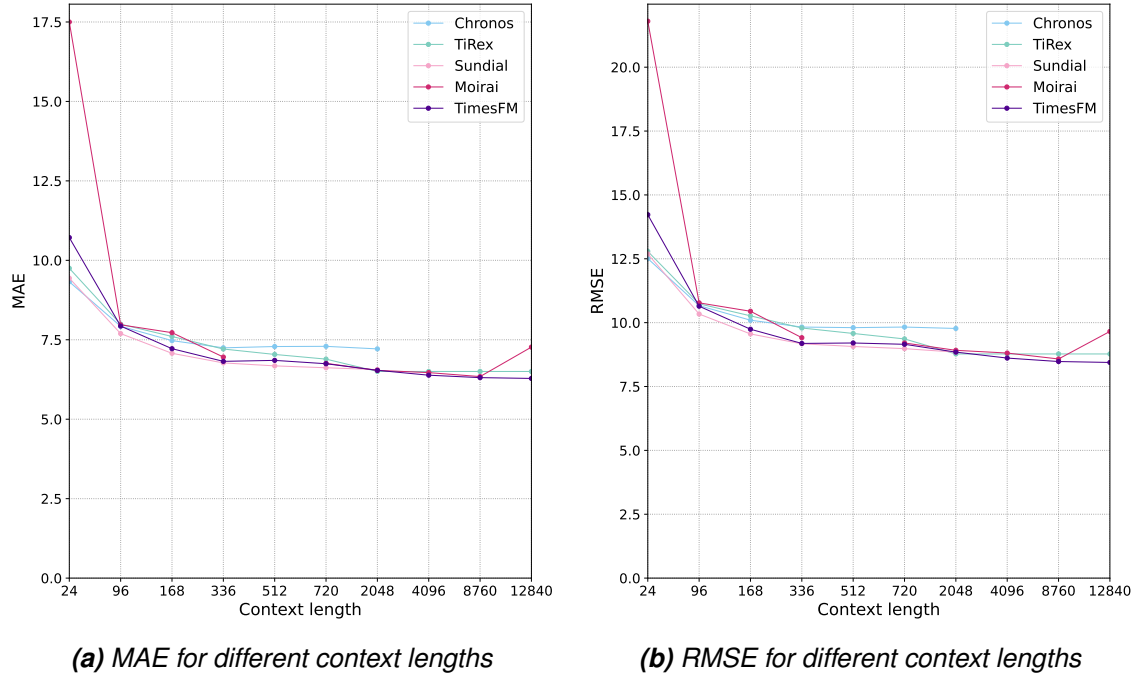


Figure 4.1. MAE and RMSE for different context lengths. All models take univariate input.

4.3 Effect of covariates

Moirai was the only model tested in this thesis which supported external covariates. Table 4.2 shows that surprisingly Moirai could not improve its performance with external covariates. Its performance deteriorated significantly.

This outcome suggested an anomaly. After investigation, a GitHub discussion [40] from Moirai's repository was found complaining about the same issue. A repository contributor stated that

current moirai2 is pre-trained with univariate data, and it does not learn any cross-variate patterns ...

This directly explains why this model version could not use external covariates, even though there is an option to give this model external covariates as input.

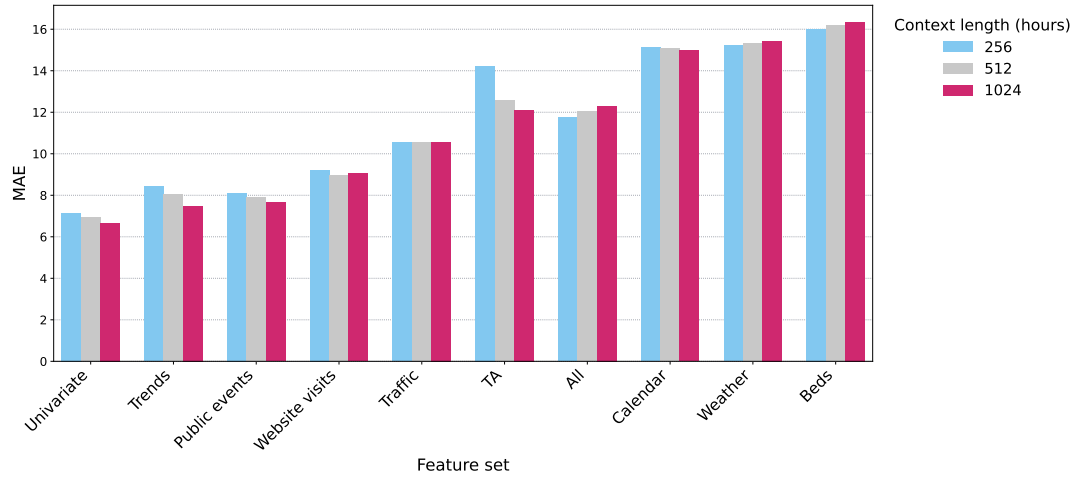


Figure 4.2. MAE across different multivariate feature sets. The "All" category includes all covariates, while other groups represent specific covariate subsets. Different colors show results for different context lengths.

4.4 Computational cost

Table 4.2 shows the computing time (latency) in seconds for each forecasting model across different context lengths. Table 4.3 presents the peak RAM usage in GiB for each model. Missing values indicate the model does not support that context length.

Table 4.3 shows that Moirai is the most memory efficient model. Table 4.2 shows that for small context lengths Chronos is the fastest model and for larger context lengths TiRex and Moirai are the fastest.

Context Length	Chronos	Sundial	TiRex	TimesFM	Moirai	Moirai (all)
24	1.5	173	104	48.2	6.5	—
96	2.4	162	107	63.9	5.2	—
168	4.0	170	91.8	65.6	5.6	—
336	6.7	169	92.9	68.8	5.8	—
512	9.5	172	94.0	70.8	7.2	110
720	13.9	189	91.5	73.9	12.3	—
2048	41.3	192	94.3	134	17.3	—
4096	—	—	103	197	20.6	—
8760	—	—	93.5	437	70.8	—
12840	—	—	98.5	635	94.6	—

Table 4.2. Computing time (latency) in seconds for each forecasting model across different context lengths. Latency represents the total time required to generate predictions for the entire test set spanning one year. "all" represent predictions made with all 167 covariates. Missing values indicate the model does not support that context length. Note that multivariate predictions were made on more powerful hardware.

Context Length	Chronos	Sundial	TiRex	TimesFM	Moirai	Moirai (all)
24	1.73	2.89	1.58	2.51	0.78	—
96	1.87	3.18	1.59	2.51	0.78	—
168	2.05	3.45	1.59	2.51	0.78	—
336	2.51	3.99	1.59	2.51	0.78	—
512	2.59	3.65	1.59	2.51	0.78	14.0
720	5.50	3.99	1.59	2.51	0.78	—
2048	5.48	6.79	1.59	2.51	0.78	—
4096	—	—	1.60	2.51	0.78	—
8760	—	—	1.62	2.51	0.78	—
12840	—	—	1.64	2.51	0.78	—

Table 4.3. Peak RAM usage in GiB for each forecasting model across different context lengths. Peak RAM represents the maximum memory consumed throughout the entire forecasting pipeline, including data preprocessing, model loading, and prediction generation. "all" represent predictions made with all 167 covariates. Missing values indicate that model does not support this length. Note that multivariate predictions from Moirai were done on more powerful hardware.

4.5 Limitations

The evaluation covers a single Finnish emergency department and one year of testing data, so the conclusions may not transfer directly to other hospitals with different arrival patterns or operational practices. Hardware configurations differed between the univariate and multivariate Moirai experiments, which complicates precise cost comparisons. Finally, only one FM (Moirai) currently accepts the available covariates. The Moirai model was also not trained with exogenous covariates. These aspects limit the generalization of the negative covariate result.

5. CONCLUSION

Emergency department crowding worsens clinical outcomes and strains staff and resources. This thesis examined whether recently released foundational time series models can forecast hourly ED occupancy accurately enough in a zero-shot setting to be useful in practice. Using Tampere University Hospital’s ED "Acuta" data and the experimental protocol aligned with prior work, five open-source FMs were compared against established baselines and assessed how context length, external covariates, and computational requirements affect their use.

First, we evaluated predictive accuracy. All univariate FMs except Chronos moderately outperformed the strongest baseline (LightGBM). TimesFM achieved the best overall accuracy, while TiRex, Sundial, and Moirai were close when given similarly long histories. Chronos underperformed, likely it being the oldest model [32]. Increasing context length consistently improved accuracy across models, with diminishing returns beyond roughly 2,048 hours. These results show that zero-shot FMs can deliver competitive accuracy for ED occupancy forecasting.

Forecasting with external covariates was tested. From the models used, only Moirai had the ability to include covariates in its prediction. Results were significantly worse than the simple seasonal naïve baseline. The model failed to use covariates, and no general conclusions can be made on the use of external covariates in ED occupancy forecasting.

Finally, we measured computational costs. Moirai was the most memory-efficient model. Chronos produced the fastest inferences at short contexts, while TiRex and Moirai were fastest at longer contexts. Inference times for all models were practical for hourly operational use on modern hardware. These findings indicate that zero-shot FMs can reduce the operational burden associated with frequent model retraining, enabling faster experimentation and easier deployment across sites.

This work has limitations. We studied a single hospital over a fixed period and evaluated one FM with multivariate inputs. Model finetuning and careful selection of covariates were out of scope. Future research should (1) replicate across multiple hospitals and time periods; (2) test effect of fine-tuning; and (3) investigate covariate selection possibly with future models that support external covariates.

In conclusion, foundational time series models already provide accurate, fast, and op-

erationally attractive zero-shot forecasts for ED occupancy. With long historical context, several models cluster in performance and surpass strong baselines, substantially lowering the barrier to adoption by removing per-site training. While multivariate zero-shot use did not help in our setting, careful covariate design and light adaptation may close that gap. Taken together, these results support zero-shot FMs as a practical starting point for ED forecasting and outline a path toward even stronger, fine-tuned and reliable deployments.

6. REFERENCES

- [1] M. Sartini, A. Carbone, A. Demartini, *et al.*, “Overcrowding in emergency department: Causes, consequences, and solutions—a narrative review”, *Healthcare*, vol. 10, no. 9, p. 1625, Aug. 25, 2022, ISSN: 2227-9032. DOI: 10.3390/healthcare10091625.
- [2] L. M. Berg, A. Ehrenberg, J. Florin, J. Östergren, A. Discacciati, and K. E. Göransson, “Associations between crowding and ten-day mortality among patients allocated lower triage acuity levels without need of acute hospital care on departure from the emergency department”, *Annals of Emergency Medicine*, vol. 74, no. 3, pp. 345–356, Sep. 1, 2019, ISSN: 0196-0644. DOI: 10.1016/j.annemergmed.2019.04.012.
- [3] S. Jo, Y. H. Jin, J. B. Lee, T. Jeong, J. Yoon, and B. Park, “Emergency department occupancy ratio is associated with increased early mortality”, *Journal of Emergency Medicine*, vol. 46, no. 2, pp. 241–249, Feb. 1, 2014, ISSN: 0736-4679. DOI: 10.1016/j.jemermed.2013.05.026.
- [4] M. L. McCarthy, S. L. Zeger, R. Ding, *et al.*, “Crowding delays treatment and lengthens emergency department length of stay, even among high-acuity patients”, *Annals of Emergency Medicine*, vol. 54, no. 4, 492–503.e4, Oct. 1, 2009, ISSN: 0196-0644. DOI: 10.1016/j.annemergmed.2009.03.006.
- [5] J. Tuominen, E. Pulkkinen, J. Peltonen, *et al.*, “Forecasting emergency department occupancy with advanced machine learning models and multivariable input”, *International Journal of Forecasting*, vol. 40, no. 4, pp. 1410–1420, Oct. 1, 2024, ISSN: 0169-2070. DOI: 10.1016/j.ijforecast.2023.12.002.
- [6] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 2nd ed. Melbourne, Australia: OTexts, 2018. [Online]. Available: <https://otexts.com/fpp2/>.
- [7] R. Bommasani, D. A. Hudson, E. Adeli, *et al.*, *On the opportunities and risks of foundation models*, 2022. arXiv: 2108.07258 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/2108.07258>.
- [8] J. Tuominen, E. Pulkkinen, J. Peltonen, *et al.*, *Ed-ml-multivar: Forecasting emergency department occupancy with advanced machine learning models and multivariable input*, <https://github.com/tuomijal/ed-ml-multivar>, Code and data for: <https://doi.org/10.1016/j.ijforecast.2023.12.002>, 2023.

- [9] M. Gul and E. Celik, "An exhaustive review and analysis on applications of statistical forecasting in hospital emergency departments", *Health Systems*, vol. 9, no. 4, pp. 263–284, 2018, ISSN: 2047-6965. DOI: 10.1080/20476965.2018.1547348.
- [10] B. R. Asplin, D. J. Magid, K. V. Rhodes, L. I. Solberg, N. Lurie, and C. A. Camargo Jr, "A conceptual model of emergency department crowding", *Annals of emergency medicine*, vol. 42, no. 2, pp. 173–180, 2003. DOI: 10.1067/mem.2003.302.
- [11] D. B. Richardson, "Increase in patient mortality at 10 days associated with emergency department overcrowding", *The Medical Journal of Australia*, vol. 184, no. 5, pp. 213–216, Mar. 6, 2006, ISSN: 0025-729X. DOI: 10.5694/j.1326-5377.2006.tb00204.x.
- [12] Canadian Journal of Emergency Medicine, "Em overcrowding: The symptom of a system in crisis", *Canadian Journal of Emergency Medicine*, vol. 3, no. 2, pp. 82–84, 2001. DOI: 10.1017/S1481803500005285.
- [13] E. B. Kulstad, R. Sikka, R. T. Sweis, K. M. Kelley, and K. H. Rzechula, "Ed overcrowding is associated with an increased frequency of medication errors", *The American Journal of Emergency Medicine*, vol. 28, no. 3, pp. 304–309, 2010, ISSN: 0735-6757. DOI: <https://doi.org/10.1016/j.ajem.2008.12.014>.
- [14] K. Bond, M. Ospina, S. Blitz, *et al.*, "Frequency, determinants and impact of overcrowding in emergency departments in canada: A national survey", *Healthcare quarterly (Toronto, Ont.)*, vol. 10, pp. 32–40, Feb. 2007. DOI: 10.12927/hcq.2007.19312.
- [15] S. L. Bernstein, D. Aronsky, R. Duseja, *et al.*, "The effect of emergency department crowding on clinically oriented outcomes", *Academic Emergency Medicine*, vol. 16, no. 1, pp. 1–10, 2009. DOI: <https://doi.org/10.1111/j.1553-2712.2008.00295.x>.
- [16] L. M. Zibners, B. K. Bonsu, J. R. Hayes, and D. M. Cohen, "Local weather effects on emergency department visits: A time series and regression analysis", *Pediatric emergency care*, vol. 22, no. 2, pp. 104–106, 2006. DOI: 10.1097/01.pec.0000199561.34475.29.
- [17] D. Golmohammadi, "Predicting hospital admissions to reduce emergency department boarding", *International Journal of Production Economics*, vol. 182, pp. 535–544, 2016, ISSN: 0925-5273. DOI: <https://doi.org/10.1016/j.ijpe.2016.09.020>.
- [18] M. Gül and A. Guneri, "Forecasting patient length of stay in an emergency department by artificial neural networks", *Journal of Aeronautics and Space Technologies (Havacilik ve Uzay Teknolojileri Dergisi)*, vol. 8, pp. 43–48, Jul. 2015. DOI: 10.7603/s40690-015-0015-7.
- [19] R. A. D. Peter J. Brockwell, *Introduction to Time Series and Forecasting*. 2016. DOI: 10.1007/978-3-319-29854-2.

- [20] G. James, D. Witten, T. Hastie, and R. Tibshirani, *An Introduction to Statistical Learning: with Applications in R*, 2nd ed. New York, NY: Springer, 2021, ISBN: 978-1-0716-1417-4. DOI: 10.1007/978-1-0716-1418-1. [Online]. Available: <https://www.statlearning.com/>.
- [21] H. K. Saravanan, S. Dwivedi, P. Praveen, and P. Arjunan, “Analyzing the performance of time series foundation models for short-term load forecasting”, in *Proceedings of the 11th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*, ser. BuildSys '24, Hangzhou, China: Association for Computing Machinery, 2024, pp. 346–349, ISBN: 9798400707063. DOI: 10.1145/3671127.3699536.
- [22] I. Arab and R. Benitez, *Benchmarking time series forecasting models: From statistical techniques to foundation models in real-world applications*, 2025. arXiv: 2502.03395 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/2502.03395>.
- [23] C. Raffel, N. Shazeer, A. Roberts, *et al.*, “Exploring the limits of transfer learning with a unified text-to-text transformer”, *The Journal of Machine Learning Research*, vol. 21, no. 140, pp. 1–67, 2020. [Online]. Available: <http://jmlr.org/papers/v21/20-074.html>.
- [24] A. Vaswani, N. Shazeer, N. Parmar, *et al.*, “Attention is all you need”, in *Advances in Neural Information Processing Systems*, 2017, pp. 5998–6008. [Online]. Available: 10.5555/3295222.3295349.
- [25] Y. LeCun, Y. Bengio, and G. Hinton, “Deep learning”, *Nature*, vol. 521, no. 7553, pp. 436–444, 2015, Published online: 27 May 2015; Issue date: 28 May 2015. DOI: 10.1038/nature14539. [Online]. Available: <https://www.nature.com/articles/nature14539>.
- [26] B. Lim, S. Ö. Arık, N. Loeff, and T. Pfister, “Temporal fusion transformers for interpretable multi-horizon time series forecasting”, *International Journal of Forecasting*, vol. 37, no. 4, pp. 1748–1764, 2021. DOI: 10.1016/j.ijforecast.2021.03.012.
- [27] P. Domingos, “A few useful things to know about machine learning”, *Communications of the ACM*, vol. 55, no. 10, pp. 78–87, 2012. DOI: 10.1145/2347736.2347755.
- [28] T. Aksu, G. Woo, J. Liu, *et al.*, *GIFT-Eval: A benchmark for general time series forecasting model evaluation*, 2024. DOI: 10.48550/arXiv.2410.10393. arXiv: 2410.10393 [cs.LG].
- [29] A. Das, W. Kong, R. Sen, and Y. Zhou, *A decoder-only foundation model for time-series forecasting*, Apr. 17, 2024. DOI: 10.48550/arXiv.2310.10688. arXiv: 2310.10688.
- [30] Salesforce AI Research, *GIFT-Eval leaderboard: General time series forecasting model evaluation*, <https://huggingface.co/spaces/Salesforce/GIFT-Eval>, Accessed: October 7, 2025, 2024.

- [31] A. F. Ansari, L. Stella, C. Turkmen, *et al.*, *Chronos: Learning the language of time series*, Nov. 4, 2024. DOI: 10.48550/arXiv.2403.07815. arXiv: 2403.07815. [Online]. Available: <http://arxiv.org/abs/2403.07815>.
- [32] Amazon Science, *Chronos-Bolt-Base Model Card*, Hugging Face Repository, Accessed: 2025-10-19, 2024. [Online]. Available: <https://huggingface.co/amazon/chronos-bolt-base>.
- [33] A. F. Ansari *et al.*, *Chronos-2: From univariate to universal forecasting*, 2025. arXiv: 2510.15821 [cs.LG]. [Online]. Available: 10.48550/arXiv.2510.15821.
- [34] Y. Liu, G. Qin, Z. Shi, *et al.*, *Sundial: A family of highly capable time series foundation models*, May 30, 2025. DOI: 10.48550/arXiv.2502.00816. arXiv: 2502.00816. [Online]. Available: <http://arxiv.org/abs/2502.00816>.
- [35] G. Woo, C. Liu, A. Kumar, C. Xiong, S. Savarese, and D. Sahoo, *Unified training of universal time series forecasting transformers*, May 22, 2024. DOI: 10.48550/arXiv.2402.02592. arXiv: 2402.02592. [Online]. Available: <http://arxiv.org/abs/2402.02592>.
- [36] I. T. Aksu, J. Liu, and Salesforce AI Research, *Introducing Moirai 2.0: A New Standard for Time Series Foundation Models*, Salesforce Engineering Blog, Accessed: 2025-09-15, Aug. 2025. [Online]. Available: <https://www.salesforce.com/blog/moirai-2-0/>.
- [37] A. Auer, P. Podest, D. Klotz, S. Böck, G. Klambauer, and S. Hochreiter, *TiRex: Zero-shot forecasting across long and short horizons with enhanced in-context learning*, May 29, 2025. DOI: 10.48550/arXiv.2505.23719. arXiv: 2505.23719. [Online]. Available: <http://arxiv.org/abs/2505.23719> (visited on 09/11/2025).
- [38] M. Beck, K. Pöppel, M. Spanring, *et al.*, “xLSTM: Extended long short-term memory”, in *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, Nov. 2024.
- [39] G. Ke, Q. Meng, T. Finley, *et al.*, “Lightgbm: A highly efficient gradient boosting decision tree”, in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 3146–3154. DOI: 10.5555/3294996.3295074. [Online]. Available: <https://proceedings.neurips.cc/paper/2017/file/6449f44a102fde848669bdd9eb6b76fa-Paper.pdf>.
- [40] O. Shchur, *Moirai2 forecast accuracy drops significantly when feat_dynamic_real are present*, <https://github.com/SalesforceAIResearch/uni2ts/issues/224>, GitHub Issue. SalesforceAIResearch/uni2ts Issue #224, Aug. 2025.

APPENDIX A: COMPUTATIONAL COSTS FOR
MULTIVARIATE FORECASTS

Feature Subset	Context Length		
	256	512	1024
All	43.3	110	370
Beds	5.3	5.6	9.9
Calendar	5.3	5.4	5.6
Events	5.2	5.2	5.4
TA	5.5	7.8	15.6
Traffic	7.4	14.1	39.2
Trends	5.1	5.2	5.3
Weather	7.3	7.5	8.2
Website	5.2	5.2	5.4

Table A.1. Computing time (latency) in seconds for different feature subsets across context lengths (Moirai model). "All" represents all covariates combined, while other columns represent individual covariate features.

Feature Subset	Context Length		
	256	512	1024
All	1030	3730	14300
Beds	68.4	108	252
Calendar	56.5	66.1	90.0
Events	53.5	53.8	54.9
TA	84.4	171	498
Traffic	146	400	1400
Trends	53.5	54.3	55.3
Weather	62.4	78.9	144
Website	53.9	55.8	64.9

Table A.2. Peak RAM usage in MiB for different feature subsets across context lengths for Moirai model. "All" represents all covariates combined, while other columns represent covariate subsets.